Forensic Evidence Collection and Preservation Guide

Project: Python-Based Automated Web Penetration Testing Toolkit
Investigator: Drew
Evidence Case ID: PENTEST-001
Purpose: Document the procedures for acquiring, preserving, verifying, and managing forensic evidence using FTK Imager, KAPE, and DumpIt.

---

1. Objective

This procedure demonstrates how digital forensic acquisition can complement the Python-based web penetration-testing toolkit.

The objective is to:

- Capture live forensic data from an authorized laboratory system.
- Create and verify a forensic disk image.
- Acquire volatile system memory.
- Calculate and validate MD5, SHA-1, and SHA-256 hashes.
- Maintain evidence integrity through appropriate write-blocking mechanisms.
- Document acquisition timestamps, tool versions, parameters, and responsible personnel.
- Integrate acquired artifacts into the project's existing evidence register and chain-of-custody system.

Current implementation status: This document describes the planned procedures. The acquisitions must be performed and their results recorded before they can be reported as completed.

2. Acquisition Environment Preparation

2.1 Required equipment and software

Component	Purpose
Windows workstation	Run Windows forensic acquisition tools
FTK Imager	Create and verify forensic disk images

KAPE	Collect Windows forensic artifacts from a running system
DumpIt	Acquire volatile Windows system memory
External evidence drive	Store acquired forensic artifacts
Hardware write blocker	Prevent writes to applicable physical source media
Python evidence framework	Register and verify collected evidence

2.2 Prepare the evidence storage location

Connect an appropriately formatted external evidence drive.

For this example:

- `C:` represents the Windows workstation.
- `E:` represents the separate evidence storage drive.

Open Command Prompt as Administrator and execute:

```
None
mkdir E:\Evidence
mkdir E:\Evidence\DiskImages
mkdir E:\Evidence\Memory
mkdir E:\Evidence\KAPE
mkdir E:\Evidence\Logs
```

Confirm that the directories were created:

```
None
dir E:\Evidence
```

Ensure that the destination has sufficient free space. Disk images may require storage approximately equal to the source disk's used or total capacity, depending on acquisition format and compression. Memory dumps can be approximately the size of installed physical memory.

2.3 Record the acquisition environment

Run:

```
None
systeminfo
```

Record:

- Windows version.
- System architecture.
- Computer name.
- Installed memory.
- System date and time.

Record the current time zone:

```
None
tzutil /g
```

Record the workstation's date and time:

```
None
Get-Date -Format o
```

Use a consistent time zone throughout the acquisition records.

2.4 Document the tools

For each acquisition tool:

1. Record its official download source.
2. Record the installed version.
3. Record the executable filename.
4. Record the acquisition date.
5. Preserve the tool's documentation or version information.

Do not assume that different releases use identical command-line options.

---

3. Stage One: Disk Image Acquisition Using FTK Imager

3.1 Purpose

FTK Imager will be used to create a forensic copy of an authorized laboratory storage device.

The resulting image will preserve the source data for subsequent examination without requiring analysis of the original source.

3.2 Obtain FTK Imager

1. Visit the official Exterro website:
   https://www.exterro.com/ftk-imager
2. Locate the FTK Imager download.
3. Download the installer.
4. Install the application using the installation wizard.
5. Launch FTK Imager.
6. Record the installed version.

3.3 Prepare the source disk

1. Identify the authorized laboratory disk to be acquired.
2. Record the disk model, capacity, serial number, and other available identifiers.
3. If acquiring physical media, connect it through a compatible hardware write blocker.
4. Confirm that the write blocker is functioning and that the source is exposed read-only.
5. Record the write-blocker manufacturer, model, serial number, and verification procedure.

For a virtual machine disk, shut down the VM cleanly and acquire an appropriately protected copy using a supported workflow. Document the virtual disk format and the read-only protections applied.

A virtual machine snapshot or backup is not automatically a hardware write blocker.

3.4 Create the forensic image

1. Open FTK Imager.
2. Select **File → Create Disk Image**.
3. In the source selection window, choose the appropriate source type.
   For a connected physical device, select the physical drive.
   For a supported mounted source, select the appropriate logical drive or source.
4. Click **Next**.
5. Select the correct source disk.
6. Confirm the device identifier and capacity.
7. Click **Finish**.
8. In the Create Image window, select **Add**.
9. Select the image type. E01 is a suitable choice for this exercise.
10. Enter the evidence information:

- Case Number: PENTEST-001
- Evidence Number: DISK-001

- Examiner: Drew
- Description: Authorized Laboratory Disk Acquisition

11. Choose the output directory:

```
E:\Evidence\DiskImages
```

12. Enter an appropriate image filename:

```
LAB_DISK_001.E01
```

13. Enable **Verify images after they are created**.
14. Enable MD5 and SHA-1 calculations where available.
15. Review the acquisition settings.
16. Click **Start**.

FTK Imager will begin acquiring the selected source and display acquisition progress.

3.5 Record the acquisition results

After the acquisition completes:

1. Review the completion message.
2. Open the image summary or acquisition log.
3. Record the image filename.
4. Record the acquisition start and completion times.
5. Record the source device information.
6. Record the MD5 and SHA-1 values.
7. Record any bad sectors or acquisition errors.
8. Save the acquisition log.

The expected output is a forensic image file and an acquisition record containing its verification information.

3.6 Independently verify the disk image

In FTK Imager:

1. Select **File** → **Add Evidence Item**.
2. Select the image file option.
3. Browse to `LAB_DISK_001.E01`.
4. Add the image to the Evidence Tree.
5. Select the image.
6. Select **File** → **Verify Drive/Image**.
7. Allow verification to finish.
8. Review the verification summary.

9. Record the computed MD5 and SHA-1 values.
10. Confirm that the stored and calculated hashes match where the image format provides stored acquisition hashes.

The verification result should demonstrate whether the image matches its original acquisition hash.

3.7 Calculate an additional SHA-256 hash

Open PowerShell:

```
None
Get-FileHash "E:\Evidence\DiskImages\LAB_DISK_001.E01" -Algorithm SHA256
```

Save the output.

Record the SHA-256 value in the acquisition log.

Note that this hashes the E01 container file. It is an additional check and is not interchangeable with FTK Imager's source-media verification.

3.8 Evidence to capture

Save screenshots showing:

- FTK Imager version.
- Source disk selection.
- Image format and acquisition settings.
- Successful image creation.
- MD5 and SHA-1 verification.
- SHA-256 calculation.

Expected deliverable: A verified disk image, acquisition log, hash records, and screenshots.

---

4. Stage Two: Live System Artifact Collection Using KAPE

4.1 Purpose

KAPE will collect selected forensic artifacts from an authorized Windows system while it is running.

Unlike a full disk image, KAPE performs targeted forensic collection. It can collect operating-system artifacts such as event logs, registry data, and other configured forensic files.

4.2 Obtain KAPE

1. Visit Kroll's official KAPE resource: https://www.kroll.com/en/services/cyber/reactive-services/kroll-artifact-parser-and-extractor-kape
2. Follow the official download instructions.
3. Download the KAPE package.
4. Extract the package into a dedicated directory, such as: `C:\Tools\KAPE`
5. Open the extracted directory.
6. Locate `kape.exe`.
7. Record the package version and download information.

KAPE is available for educational and research use under Kroll's stated usage terms.

4.3 Identify collection targets

KAPE uses Targets to identify which files and artifacts should be collected.

Open Command Prompt and navigate to:

```
None
cd C:\Tools\KAPE
```

Display available targets using:

```
None
kape.exe --tlist
```

Review the available configurations.

For a general Windows forensic triage collection, look for the `!SANS_Triage` target if it is included in the installed configuration.

You can inspect its details and required collection paths using the target documentation.

Do not assume every KAPE installation contains the same targets.

4.4 Configure the collection destination

Create a separate collection directory:

```
None
mkdir E:\Evidence\KAPE
```

The source system will be the running Windows workstation.

The destination will be the external evidence drive.

4.5 Execute the collection

Open Command Prompt as Administrator.

Navigate to the KAPE installation directory:

```
None
cd C:\Tools\KAPE
```

Execute the following example, after confirming that the target exists:

```
None
kape.exe --tsource C: --tdest E:\Evidence\KAPE --target
!SANS_Triage
```

Parameter explanations:

- `--tsource C:` identifies the source volume.
- `--tdest E:\Evidence\KAPE` specifies the collection destination.
- `--target !SANS_Triage` selects the configured forensic triage collection.

KAPE will inspect the selected source and collect the artifacts specified by the target configuration.

4.6 Review the results

After collection:

1. Navigate to `E:\Evidence\KAPE`.
2. Locate the generated collection directory.
3. Review the output files.
4. Inspect the KAPE execution log.

5. Record the number of collected files.
6. Record the total collection size.
7. Document any inaccessible or skipped files.
8. Capture screenshots of the successful collection.

Do not treat a successful command execution as proof that every requested artifact was collected. Review the logs and output.

4.7 Calculate collection hashes

For a file-level manifest of the collection, run:

```
None
Get-ChildItem "E:\Evidence\KAPE" -File -Recurse |
Get-FileHash -Algorithm SHA256 |
Export-Csv "E:\Evidence\Logs\KAPE_SHA256.csv" -NoTypeInformation
```

This generates a SHA-256 manifest of the files found under the collection directory.

Review the manifest and preserve it with the acquisition records.

4.8 Register the collection

Assign:

- Evidence ID: LIVE-001
- Test ID: FORENSICS-003
- Evidence Type: Live System Triage
- Tool: KAPE
- Source: Authorized Windows Laboratory System

Record the actual timestamps, version, target configuration, destination, and manifest location.

Expected deliverable: KAPE collection output, execution log, SHA-256 manifest, and screenshots.

---

5. Stage Three: Memory Acquisition Using DumpIt

5.1 Purpose

DumpIt will acquire volatile memory from an authorized running Windows laboratory system.

Volatile memory can contain information about running processes and other system activity that may be unavailable after shutdown.

5.2 Obtain DumpIt

1. Visit Comae Technologies' official resource: https://www.comae.com/
2. Locate the DumpIt/Comae Toolkit resources.
3. Obtain an appropriate version.
4. Review the license and registration requirements.
5. Extract the package to a dedicated directory or prepared evidence USB drive.
6. Record the version and supported architecture.

Confirm that the executable is compatible with the target Windows system.

5.3 Prepare the memory acquisition

1. Boot the authorized Windows laboratory system.
2. Confirm that it is running normally.
3. Connect the prepared evidence destination.
4. Confirm that adequate storage is available.
5. Record the system date and time.
6. Record the Windows version and installed memory.
7. Record the investigator identification.

Memory acquisition is inherently a live operation. Document that the acquisition tool itself may alter some system state.

5.4 Run DumpIt

Open Command Prompt as Administrator.

Navigate to the appropriate DumpIt executable directory.

For example:

```
None
cd Z:\comae\x64
```

The drive letter and architecture directory must match the actual installation.

An example DumpIt command is:

```
None
DumpIt.exe /OUTPUT E:\Evidence\Memory\MEMORY_001.bin
```

Some versions prompt the operator to confirm the acquisition.

Follow the prompt and authorize the collection.

The tool should begin acquiring physical memory and writing the output file.

The exact command syntax should be checked against the documentation supplied with the installed release.

5.5 Confirm the acquisition

After the acquisition completes:

1. Navigate to `E:\Evidence\Memory`.
2. Confirm that `MEMORY_001.bin` exists.
3. Record the file size.
4. Record the acquisition start time.
5. Record the acquisition completion time.
6. Record any tool messages or errors.
7. Capture the completion screen.

If the acquisition fails, document the failure rather than registering the artifact as successfully acquired.

5.6 Calculate the memory dump hash

Open PowerShell:

```
None
Get-FileHash "E:\Evidence\Memory\MEMORY_001.bin" -Algorithm SHA256
```

Record the output.

If DumpIt provides an acquisition hash, record its algorithm and value separately.

The SHA-256 hash should be recalculated after transferring the artifact to evidence storage and compared against the original reference value.

5.7 Register the memory evidence

Use:

- Evidence ID: MEM-001
- Test ID: FORENSICS-004
- Evidence Type: Physical Memory Dump
- Tool: DumpIt
- Source: Authorized Windows Laboratory System

Record the acquisition metadata and verification result.

Expected deliverable: Memory dump, SHA-256 record, acquisition details, and screenshots.

---

6. Stage Four: Evidence Integrity and Write-Blocking

6.1 Purpose

This stage demonstrates how the collected artifacts are protected against unauthorized modification and how their integrity is checked.

6.2 Write-blocking procedure

For physical disk acquisition:

1. Identify the source storage device.
2. Connect it through a compatible hardware write blocker.
3. Verify that the write blocker is operating correctly.
4. Record the hardware model and serial number.
5. Document the read-only state.
6. Perform acquisition through FTK Imager.
7. Preserve the original source and acquired image.

Do not claim write-blocking merely because the acquisition tool was run with administrator privileges or because the source was mounted read-only.

For virtual disks, document the specific read-only protection applied to the source copy and explain that hardware write-blocking was not applicable.

For memory acquisition, explain that traditional disk write blockers do not prevent all changes to live system memory. Record the acquisition method and its operational limitations.

6.3 Verify transferred artifacts

After transferring evidence to its final storage location, calculate its hash again.

For example:

```
None
Get-FileHash "E:\Evidence\DiskImages\LAB_DISK_001.E01" -Algorithm SHA256
```

Compare the result with the original recorded hash.

Repeat for the memory dump and collected KAPE files.

Record each comparison as:

- PASS: The hashes match.
- FAIL: The hashes differ.
- NOT PERFORMED: No comparison was conducted.

6.4 Preserve verification records

Save:

- FTK Imager verification results.
- Original SHA-256 values.
- Post-transfer SHA-256 values.
- KAPE file manifest.
- Memory dump hash record.
- Any failed verification attempts.

Expected deliverable: Documented integrity verification for each acquired artifact.

---

7. Stage Five: Integrate Forensic Evidence Into the Python Framework

7.1 Purpose

The existing Python framework already contains an evidence register, chain-of-custody logging, SHA-256 verification, and automated audit tests.

The objective is to reuse those components for forensic acquisition artifacts.

7.2 Add an acquisition log

Create:

`evidence/forensics/logs/acquisition_log.csv`

Use these columns:

```
None
Acquisition_ID,Evidence_ID,Acquisition_Type,Source,Tool_Name,Tool_Version,Acquisition_Start,Acquisition_End,Parameters,Write_Blocking,Hash_Algorithm,Hash_Value,Verification_Result,Collected_By,Notes
```

7.3 Register the acquired artifacts

Add three records to the existing evidence register:

Evidence ID	Evidence type	Description
DISK-001	Disk Image	FTK Imager disk acquisition
LIVE-001	Live Collection	KAPE forensic triage collection
MEM-001	Memory Dump	DumpIt memory acquisition

Record actual filenames, timestamps, hash values, and collector information.

These are forensic evidence records and do not need to be assigned vulnerability IDs such as V-001 or V-002.

7.4 Update the chain-of-custody log

Record:

1. Initial collection.
2. Initial hash verification.
3. Evidence storage.
4. Every subsequent transfer.
5. Post-transfer hash verification.
6. Investigator responsible for each event.

Preserve previous entries and append new events rather than overwriting the chain-of-custody history.

7.5 Extend automated verification

Create:

`tests/test_forensic_acquisition.py`

Include tests for:

- Unique forensic evidence IDs.
- Existing acquisition files.
- Recorded tool versions.
- Acquisition timestamps.
- SHA-256 integrity checks.
- Detection of altered evidence.
- Detection of missing evidence.
- Recorded collection handlers.
- Chain-of-custody entries.

Run the existing and new audit tests:

```
None
python -m unittest tests.test_evidence_audit
tests.test_forensic_acquisition -v
```

The tests should verify that forensic artifacts are managed consistently with the framework's existing evidence-handling procedures.

7.6 Include forensic evidence in the HTML report

Extend the report generator to display a forensic acquisition summary.

Include:

- Evidence ID.
- Acquisition type.
- Acquisition tool.
- Collection timestamp.
- Verification status.
- Hash algorithm.
- Reference to the acquisition record.

Do not embed large disk images or memory dumps directly into the HTML report. Reference their controlled storage locations and supporting hash records.

---

8. Stage Six: Final Documentation and Submission

8.1 Required files

The completed submission should contain:

```
None
evidence/
├── evidence_register.csv
├── forensics/
│   ├── FORENSIC_ACQUISITION_REPORT.md
│   ├── disk_images/
│   │   └── LAB_DISK_001.E01
│   ├── memory_dumps/
│   │   └── MEMORY_001.bin
│   ├── KAPE/
│   ├── screenshots/
│   └── logs/
│       ├── acquisition_log.csv
│       └── KAPE_SHA256.csv
└── logs/
    ├── chain_of_custody.csv
    └── verification_log.csv
```

8.2 Required screenshots

FTK Imager

- Application version.
- Source selection.
- Acquisition configuration.
- Completed image.
- MD5/SHA-1 verification.

KAPE

- Tool version.
- Selected target.
- Collection command.
- Successful collection output.
- Generated artifact directory.

DumpIt

- Tool version.
- Acquisition command.
- Completed memory dump.
- Memory dump hash.

Python framework

- Updated evidence register.
- Acquisition log.
- Chain-of-custody log.
- Verification log.
- Successful audit test results.

8.3 Final results table

Complete the following table using actual acquisition results:

Acquisition	Tool	Evidence ID	Integrity result	Status
Disk image	FTK Imager	DISK-001	Actual MD5/SHA-1 comparison	Pending
Live collection	KAPE	LIVE-001	Actual SHA-256 manifest checks	Pending
Memory dump	DumpIt	MEM-001	Actual SHA-256 comparison	Pending

8.4 Conclusion

The forensic acquisition workflow extends the Python-Based Automated Web Penetration Testing Framework by introducing additional evidence types and documenting their collection and preservation.

FTK Imager provides disk acquisition and verification, KAPE supports targeted collection of live Windows forensic artifacts, and DumpIt supports volatile memory acquisition.

The Python framework serves as the central evidence-management component, maintaining unique identifiers, collection metadata, hash records, verification results, and chain-of-custody history.

Together, these capabilities demonstrate how web penetration-testing evidence and digital forensic artifacts can be managed through a consistent, auditable evidence-handling process.

---

9. Official References

1. Exterro, FTK Imager: https://www.exterro.com/ftk-imager
2. Exterro, FTK Imager User Guide: https://d1kpmuwb7gvu1i.cloudfront.net/Imager/8.2/Exterro%20FTK%20Imager%208.2%20-%20User%20Guide-.pdf
3. Kroll, KAPE: https://www.kroll.com/en/services/cyber/reactive-services/kroll-artifact-parser-and-extractor-kape
4. Kroll, KAPE Resources: https://www.kroll.com/en/services/cyber/reactive-services/kroll-artifact-parser-and-extractor-kape/resources
5. Comae Technologies: https://www.comae.com/

Documentation note: This guide is a procedural plan, not a record of completed acquisitions. Actual tool versions, acquisition parameters, timestamps, filenames, hash values, write-blocking details, and verification results must be entered after the corresponding operations have been performed.